Des. Codes Cryptogr. (2008) 46:243–259
DOI 10.1007/s10623-007-9132-z

# Maximal caps in AG (6, 3)

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Received: 17 November 2006 / Revised: 15 August 2007 / Accepted: 4 September 2007 /
Published online: 10 January 2008


**Abstract** We show that there are no complete 44-caps in AG(5, 3). We then use this result to prove that the maximal size for a cap in AG(6, 3) is equal to 112, and that the 112-caps in AG(6, 3) are unique up to affine equivalence.

**Keywords** Finite geometry · Coding theory · Caps

**AMS Classifications** 51E22 · 51E20 · 05B25

## 1 Introduction

The main problem in the study of caps is the determination of the size of the largest caps in AG$(n, q)$ and in PG$(n, q)$.

Presently, the following exact values are known. In AG$(2, q)$ and PG$(2, q)$, $q$ odd, there are at most $(q+1)$-caps [3]. In AG$(2, q)$ and PG$(2, q)$, $q$ even, there are at most $(q+2)$-caps [3]. In AG$(3, q)$, $q > 2$, the maximal size of a cap is $q^2$, and in PG$(3, q)$, $q > 2$, the maximal size of a cap is $q^2 + 1$ [3,13]. In AG$(n, 2)$ and in PG$(n, 2)$, the maximal size of a cap is $2n$ [3].

In some cases, a complete characterization is known. Namely, in AG$(2, q)$ and in PG$(2, q)$, $q$ odd, every $(q+1)$-cap is a conic [14,15]. In AG$(2, q)$ and PG$(2, q)$, $q$ even, $q \geq 16$, distinct types of $(q + 2)$-caps exist; see [10] for a list of the known infinite classes of $(q + 2)$-caps. In PG$(3, q)$, $q$ odd, every $(q^2 + 1)$-cap is an elliptic quadric and in AG$(3, q)$, $q$ odd, every $q^2$-cap is an elliptic quadric minus one point [1,11]. In PG$(3, q)$, $q = 2^h$, $h$ odd, $h \geq 3$, next to the elliptic quadric, at least one other type of $(q^2 + 1)$-cap exists, called the *Tits ovoid*

Communicated by J.D. Key.

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[17]. In AG$(3, q)$, $q$ even, $q > 2$, every $q^2$-cap is obtained by deleting one point from a $(q^2 + 1)$-cap in PG$(3, q)$. In PG$(n, 2)$, every $2^n$-cap is the complement of a hyperplane [16].

Also some sporadic results are known. Namely, the maximal size of a cap in AG(4, 3) and in PG(4, 3) is 20 [12]. The maximal size of a cap in PG(5, 3) is 56 [7]. The maximal size of a cap in PG(4, 4) is 41 [5].

Regarding the characterizations, exactly one type of 20-cap exists in AG(4, 3) and exactly 9 types of 20-caps exist in PG(4, 3) [9]. The 56-cap in PG(5, 3) is projectively unique [8]. And there are at least 2 distinct types of 41-caps in PG(4, 4) [5].

The standard definition of a cap is a set of points in AG$(n, q)$ or PG$(n, q)$ with no three collinear. In this article, we are working with caps in AG$(n, 3)$, so we define a $k$-cap $S$ in $F_3^n$ to be a collection of vectors such that for any three distinct vectors $v_1, v_2, v_3$, we have $v_1 + v_2 + v_3 \neq 0$. The vectors in the $k$-cap form the columns of an $n$ by $k$ matrix which we denote by $M(S)$. A hyperplane is represented by a nonzero vector $h$ (equivalently $-h$) in $F_3^n$; it determines three hyperplane cosets (flats), and two vectors $v$, $w$ are in the same flat if $h^T w = h^T v$. The entries of the vector $h^T M(S)$ specify how the vectors in the cap $S$ are distributed among the flats determined by the hyperplane $h$.

We call a point that is not in the cap an exterior point.

Let $d(n) = (3^n - 1)/2$ be the number of distinct lines through the origin, or equivalently, the number of hyperplanes in $F_3^n$.

Given a cap $S$, let $a_h(S) = a_h$, $b_h(S) = b_h$, $c_h(S) = c_h$ be the number of vectors $v$ in $S$ for which $h^T v = -1, 0, 1$ respectively. Define the division $x$-$y$-$z$ corresponding to $h$ by $x = \max(a_h, b_h, c_h)$, $y = \text{middle}(a_h, b_h, c_h)$, $z = \min(a_h, b_h, c_h)$.

Define the pair count

$$N_2(h) = N_2 = \frac{a_h(a_h - 1)}{2} + \frac{b_h(b_h - 1)}{2} + \frac{c_h(c_h - 1)}{2}.$$

This is the number of pairs of points that are contained in the same flat for a given hyperplane. Define the triple count

$$N_3(h) = N_3 = \frac{a_h(a_h - 1)(a_h - 2)}{6} + \frac{b_h(b_h - 1)(b_h - 2)}{6} + \frac{c_h(c_h - 1)(c_h - 2)}{6}.$$

This is the number of triples that are contained in the same flat for a given hyperplane.

We will prove that there are no complete 44-caps in AG(5, 3) (Theorem 14) and we will prove that the maximal size for a cap in AG(6, 3) is equal to 112, and that the 112-caps in AG(6, 3) are unique up to affine equivalence. In Theorem 16, we prove that every 112-cap in AG(6, 3) is equal to a cone with vertex $c$ and base a 56-cap in PG(5, 3), except for the vertex $c$ and for the points of the cone in one hyperplane not through $c$.

## 2 Preliminary results on 112-caps in AG(6, 3)

We begin our analysis of 112-caps by showing that any such cap must admit one of several divisions. The 45-45-22 division is associated with the classical 112-cap, and we will show that each of the other divisions can be eliminated using assumptions about 44-caps in AG(5,3).

**Theorem 1** *Any 112-cap in AG*(6, 3) *must have a* 45-45-22 *division, a* 45-44-23 *division, a* 44-44-24 *division, or a* 45-43-24 *division.*

*Proof* We prove this theorem in a sequence of lemmas by applying the combinatorial techniques found in [4, pp. 9–12].

We will make use of the following lemmas, which can be proved using simple counting arguments.

**Lemma 2** *For a k-cap in AG(n,3),*

$$\sum_h N_2(h) = \frac{k(k-1)}{2} d(n-1).$$

**Lemma 3** *For a k-cap in AG(n,3),*

$$\sum_h N_3(h) = \frac{k(k-1)(k-2)}{6} d(n-2).$$

Rearranging the terms gives:

$$\sum_h (N_2(h) - \frac{k(k-1)}{2d(n)} d(n-1)) = 0 \quad \text{and}$$
$$\sum_h (N_3(h) - \frac{k(k-1)(k-2)}{6d(n)} d(n-2)) = 0.$$

This implies that for all $a$ and $b$, there must exist an $h$ such that:

$$a\left(N_2(h) - \frac{k(k-1)}{2d(n)} d(n-1)\right) - b\left(N_3(h) - \frac{k(k-1)(k-2)}{6d(n)} d(n-2)\right) \geq 0. \quad (2.1)$$

This equation is valid for any k-cap in AG($n$,3). We apply it to a 112-cap in 6 dimensions. Setting $a = 377$ and $b = 11$, it can be shown using a computer search that the 112-cap must admit one of the following divisions:

1. A division where $x \geq 46$.
2. A 45-45-22 division.
3. A 45-44-23 division.
4. A 44-44-24 division.
5. A 45-43-24 division.

The first possibility does not occur, since it follows from [6] that there are no 46-caps in AG(5,3). This completes the proof of Theorem 1. □

*Remark 4* The method leading to Theorem 5 in [2] yields an alternative proof of Theorem 1 that does not require a computer search.

## 3 Study of the divisions

We begin with a few facts about the 45-cap in AG(5,3). It follows from [6] that any 45-cap in AG(5,3) has 55 18-18-9 divisions and 66 15-15-15 divisions.

**Lemma 5** *For an n-cap in AG($k$, 3), consider a point p. Let* $s = \sum_{i=1}^{d(k)} s_i$*, where* $s_i$ *is the number of points of the n-cap in the same hyperplane coset as p.*

$$\textit{If } p \textit{ is in the cap}, \quad s = (n-1) \cdot d(k-1) + d(k).$$
$$\textit{If } p \textit{ is not in the cap}, \quad s = n \cdot d(k-1).$$

*Proof* If $p$ is not in the cap, for each of the $n$ points of the $n$-cap, there are $d(k-1)$ hyperplane cosets containing both $p$ and that point.

If $p$ is in the cap, there is one point in the cap, namely $p$, which is always in the same hyperplane coset as $p$. For each of the other $n-1$ points of the $n$-cap, there are $d(k-1)$ hyperplane cosets containing both $p$ and that point. □

Applying this to the 45-cap, we obtain:

**Lemma 6** *Each exterior point appears in* 20 *of the* 55 *flats meeting the* 45*-cap in* $AG(5,3)$ *in* 9 *points.*

*Proof* Let $x$ be the number of times each exterior point appears in a flat meeting the cap in 9 points.

Using Lemma 5, $18(55-x)+9x+66\cdot 15 = 1980-9x = 45\cdot 40 = 1800$, so $x=20$. □

**Lemma 7** *Each point in the* 45*-cap appears in* 11 *of the* 55 *flats meeting the* 45*-cap in* $AG(5,3)$ *in* 9 *points.*

*Proof* Let $x$ be the number of times each point appears in a flat meeting the cap in 9 points.

Using Lemma 5, $18(55-x)+9x+66\cdot 15 = 1980-9x = 44\cdot 40+121 = 1881$, so $x=11$. □

**Theorem 8** *If there is a* 45-44-23 *division for a* 112*-cap in* $AG(6,3)$, *then there is a complete* 44*-cap in* $AG(5,3)$.

*Proof* Now let us use Lemma 6 to look at 45-$m$-$r$ divisions in AG(6, 3).

For each point $v1$ in the flat with $m$ points and point $v2$ in the flat with $r$ points, let $v3=-v1-v2$.

Let us call each of these $v3$ a phantom point. There are $m\cdot r$ phantom points.

Each of these phantom points must be an exterior point to the 45-cap.

To make things easy, start with the 45-cap in the space AG(5,3). In other words, if $v$ is in the 45-cap, $v$ is of the form $(v_1,\ldots,v_5,v_6=0)$. Similarly, if $v$ is in the flat meeting the cap in $m$ points, $v_6=1$, and if $v$ is in the flat meeting the cap in $r$ points, $v_6=2$.

Now look within the 45-cap. Let $h=(h_1,\ldots,h_5)$ be a hyperplane of the 45-cap that divides it into a 9-point flat and two 18-point flats. Let $v=(v_1,v_2,v_3,v_4,v_5)$ be a vector in the flat with 9 points. We can subtract $(v_1,\ldots,v_5,\ v_6=0)$ from every vector in the cap in AG(6,3) and it will still be a cap. After doing this, returning to within the 45-cap, let $u$ and $w$ be two vectors in the original cap. Let $u'=u-v$ and $w'=w-v$. If $hu=hw$, $h(u')=hu-hv=hw-hv=hw'$. If $u$ and $w$ were in the same flat in the old cap, $u'$ and $w'$ will be in the same flat for the new cap. But $v'=v-v=0$, so if $w$ was in the 9-point flat of the old cap, $hw'=hv'=0$.

Returning to the cap in AG(6,3), form a hyperplane $\tilde{h}=(h_1,\ldots,h_5,c)$, where we will pick $c$ later. Here, $\tilde{h}$ is a hyperplane of the cap in AG(6,3) that subdivides the 45-cap in the left hyperplane coset into an 18-18-9 division with the 9-solid on top. Define $N(a,b,c)$ = number of points $v$ in the cap such that $v_6=a$, $\tilde{h}\cdot v=b$. For a given $c$, we can represent this graphically as

| | | |
|---|---|---|
| $N(0,0,c)$ | $N(1,0,c)$ | $N(2,0,c)$ |
| $N(0,1,c)$ | $N(1,1,c)$ | $N(2,1,c)$ |
| $N(0,2,c)$ | $N(1,2,c)$ | $N(2,2,c)$ |

For all $c$, $N(0,0,c)=9$, $N(0,1,c)=18$, and $N(0,2,c)=18$. Furthermore, $N(1,b,c)=N(1,b-c,0)$ because if $v_6=1$ and $(h_1,\ldots,h_5,0)\cdot v=b-c$, $(h_1,h_2,h_3,h_4,h_5,c)\cdot v=b$. We can pick $c$ so that $N(1,0,c)\geq N(1,1,c)$ and $N(1,0,c)\geq N(1,2,c)$. Finally, if $N(1,2,c)\geq N(1,1,c)$, take $\tilde{h}'=-\tilde{h}$. Using this hyperplane switches $N(a,1,c)$ with $N(a,2,c)$ for all $a$, so $N(1,1,c)\geq N(1,2,c)$ while everything else we noted is preserved.

Candidate values for the counts $N(a,b,c)$ are shown below, where $x\geq y\geq z, x+y+z=m$ and $a+b+c=r$.

| | | |
|---|---|---|
| 9 | $x$ | $a$ |
| 18 | $y$ | $b$ |
| 18 | $z$ | $c$ |

Define $f_h=xa+yc+zb$ to be the number of phantom points in the 9-point flat of the 45-cap.

Using Lemma 6, we obtain

$$\sum_h f_h=20mr, \tag{3.1}$$

where the sum is taken over the 55 18-18-9 subdivisions of the 45-point hyperplane coset.

Since the average value of $f_h$ is $\frac{20}{55}mr$, there must be at least one $h$ such that $f_h\geq\frac{20}{55}mr$.

We consider the case $m=44$ and $r=23$. There must be an $h$ such that

$$f_h\geq\frac{20}{55}\cdot(mr)=44\cdot 23\cdot\frac{20}{55}=368.$$

Recall that all 45-caps in AG(5,3) have only 18-18-9 divisions and 15-15-15 divisions, and that there are no 46-caps in AG(5,3). Note that three entries $d, e, f$ in the same row or column constitute a division of a flat, and the same is true of entries in distinct rows and columns.

Assume that there are no complete 44-caps in AG(5,3), so that every 44-cap is obtained by deleting a point from a 45-cap.

The possible cases are as follows:

1. $x=18, y=18$, and $z=8$.

| | | |
|---|---|---|
| 9 | 18 | $\leq 9$ |
| 18 | 18 | $\geq 5$ |
| 18 | 8 | $\leq 9$ |

We have $f_h=18a+18c+8b=18(a+b+c)-10b=414-10b$, and since $b\geq 5$, it follows that $f_h\leq 414-50=364$, and this case cannot give $f_h\geq 368$.

2. $x=18$, $y=17$, and $z=9$.

| | | |
|---|---|---|
| 9 | 18 | $\leq 9$ |
| 18 | 17 | $\geq 5$ |
| 18 | 9 | $\leq 9$ |

We have $f_h=18a+17c+9b=17(a+b+c)+a-8b\leq 391+9-40=360$, so this case cannot give $f_h\geq 368$.

3. $x=15$, $y=15$, and $z=14$.
Then $f_h\leq 15\cdot 23=345$, and this case cannot give $f_h\geq 368$.

If there are no complete 44-caps, there cannot be a 112-cap with a 45-44-23 division. □

*Remark 9* A similar technique can be used to show that no 44-caps have a division of the form 19-?-? or 20-?-?, as will be shown later in Theorem 13.

**Theorem 10** *If there is a 45-43-24 division for a 112-cap in AG(6,3), then there is a complete 44-cap in AG(5,3).*

*Proof* We assume that there are no complete 44-caps in AG(5,3). The average value of $f_h$ is $\frac{20}{55}mr = \frac{4128}{11} = 375.273$.

However, as will be shown, this is not sufficient to exclude this division, so we need another condition. Let $p$ be the point described by Lemma 15 in the middle flat. This point may or may not be in the cap.

Now consider the points in the left hyperplane coset that are collinear with $p$ and a point in the right flat that is in the cap. These points may or may not be in the cap. However, from Lemma 6 and Lemma 7, a point that is in the 45-cap appears 11 times in the 9-point subflat, and a point that is not in the 45-cap appears 20 times in the 9-point subflat. In either case, these points appear at most 20 times in the 9-point subflats. Let $g_h$ be the number of these points in the 9-point subflat of the left flat for a given subdivision. We have $g_{avg} \leq \frac{20}{55} \cdot 24 = \frac{96}{11}$.

Applying the same transformations as before, the counts $N(a, b, c)$ take the form:

| | | |
|---|---|---|
| 9 | $x$ | $a$ |
| 18 | $y$ | $b$ |
| 18 | $z$ | $c$ |

where $x \geq y \geq z$, $x + y + z = 43$, and $a + b + c = 24$.

First we will consider the cases when $x \leq 18$.

Assume $a \leq 2$, $b \leq 2$, or $c \leq 2$. Then $a \geq 11$, $b \geq 11$, or $c \geq 11$. Now $x \geq 15$, so the only possibility is $a \geq 11$. However, if $a = 11$, we need $b \geq 11$ or $c \geq 11$, which gives a contradiction. Also, if $a \geq 13$, $y \leq 12$ and $z \leq 12$, giving $x \geq 19$, which is a contradiction. Thus, $a = 12$. We need $b = 10$ or $c = 10$. However, in either case, we need $x = 15$. But then $y \geq 14$, which is a contradiction.

Thus, $a \geq 3$, $b \geq 3$, and $c \geq 3$, and $g_h \geq 3$.

Looking at $f_h$, the possible cases are as follows:

1. $x = 15$.
   We have $f_h \leq 15 \cdot 24 = 360$.
2. $x \geq 16$ and $y \geq 16$.
   We have $a \leq 9$, $c \leq 9$, and $b \geq 6$. Maximizing $a$ and $c$ maximizes $f_h$, so $f_h \leq 9 \cdot x + 9 \cdot y + 6 \cdot z = 9 \cdot 43 - 3 \cdot z = 387 - 3 \cdot z$. However, $z \geq 7$, and so $f_h \leq 366$.
3. $x \geq 16$ and $y \leq 15$.
   We have $a \leq 25 - y$, $c \leq 9$, and $b \geq y - 10$, for otherwise either a complete 44-cap or an inadmissable cap is created.
   Maximizing $a$ and $c$ maximizes $f_h$ and $z = 43 - x - y$, so $f_h \leq 25 \cdot x - x \cdot y + 9 \cdot y + 43 \cdot y - 430 - x \cdot y + 10 \cdot x - y^2 + 10 \cdot y = (35 - 2 \cdot y) \cdot x + 62 \cdot y - y^2 - 430$. But $y \leq 15$, so no matter what $y$ is, we want to choose $x$ as high as possible, that is, $x = 18$. So $f_h \leq 200 + 26 \cdot y - y^2$. This is maximized when $y = 13$, so $f_h \leq 369$.

In all of these cases, $f_h \leq 369$.

The possible cases when $x \geq 19$ are as follows:

1. $x = 20$.
   We have $b \leq 5$ and $c \leq 5$, so $a \geq 14$. But then $y \leq 11$ and $z \leq 11$, so $x + y + z \leq 42$. Contradiction.

2. $x = 19$ and $a \geq 14$.
   We have $y \leq 11$ and $z \leq 11$, so $x + y + z \leq 41$. Contradiction.
3. $x = 19$ and $a = 13$.
   We have $y = z = 12$ and $b = 6$, $c = 5$, or $b = 5$, $c = 6$. However, as shown before, exchanging two rows is a valid affine transformation, so both of these possibilities are equivalent to

| | | |
|---|---|---|
| 9 | 19 | 13 |
| 18 | 12 | 6 |
| 18 | 12 | 5 |

   with $f_h = 19 \cdot 13 + 12 \cdot 11 = 379$. Since the point $p$ lies in the 19-solid, we have $g_h = 13$. Call this intersection square $I$.
4. $x = 19$ and $a = 12$.
   We have $b = c = 6$, and $y$ can be either 12 or 13. Thus we get the following possibilities,

| | | |
|---|---|---|
| 9 | 19 | 12 |
| 18 | 12 or 13 | 6 |
| 18 | 12 or 11 | 6 |

   with $f_h = 19 \cdot 12 + 24 \cdot 6 = 372$ and $g_h = 12$ or 6, respectively. Call these intersection squares $I_2$ and $I_3$, respectively.
5. $x = 19$ and $a \leq 11$.
   We have $b \leq 6$ and $c \leq 6$, so $a \geq 12$. Contradiction.

We need a linear combination of intersection squares such that $f_{avg} = \frac{4128}{11}$ and $g_{avg} \leq \frac{96}{11}$. There is only one intersection square with $f_h > \frac{4128}{11}$, so the number of these intersection squares is completely determined by the others. Creating a basis for the space with $f_{avg} = \frac{4128}{11}$,

1) if only the intersection squares $I$, $I_2$ and $I_3$ occur, we have $379 - \frac{4128}{11} = \frac{41}{11}$ of $I_2$ or $I_3$, and $\frac{4128}{11} - 372 = \frac{36}{11}$ of $I$. This gives $g_{avg} \geq \frac{\frac{36}{11} \cdot 13 + \frac{41}{11} \cdot 6}{7} = \frac{102}{11}$.

2) if some intersection squares different from $I$, $I_2$ and $I_3$ occur, we have $379 - \frac{4128}{11} = \frac{41}{11}$ of some other intersection squares, and at least $\frac{4128}{11} - 369 = \frac{69}{11}$ of $I$. This gives $g_{avg} \geq \frac{\frac{69}{11} \cdot 13 + \frac{41}{11} \cdot 3}{10} = \frac{102}{11}$.

Thus, there are no such combinations.

Therefore, if there are no complete 44-caps, there are no 112-caps in AG(6,3) that have a 45-43-24 division.

**Theorem 11** *If there is a* 44-44-24 *division for a* 112*-cap in AG*(6, 3)*, then there is a complete* 44*-cap in AG*(5, 3).

*Proof* Assume that there are no complete 44-caps in AG(5,3), so that every 44-cap is obtained by deleting a point from a 45-cap.

Without loss of generality, suppose that the 45th point is added to the left hyperplane (in the 44-44-24 division). Next we count the number of times a phantom point is in a 9-point flat. From Lemma 6, if the phantom point is an exterior point, then it appears 20 times in a 9-point flat.

If the phantom point coincides with the point extending the 44-cap, by Lemma 7, it appears 11 times in a 9-point flat. Let $x$ be the number of phantom points which coincide with the point extending the 44-cap and let $N$ be the total number of times a phantom point is in a 9-point flat.

Then $N = (44 \cdot 24 - x) \cdot 20 + 11 \cdot x = 44 \cdot 24 \cdot 20 - 9 \cdot x$. Now $x \leq 24$, so $N \geq 20904$, and there must be at least one 8 or 9-point flat of the 44-cap that has at least $\frac{N}{55} > 380.07$ phantom points.

Let $h$ be the hyperplane of the 44-cap that divides it into this flat and two 17 or 18-point cosets.

Applying the transformations described above, the counts $N(a, b, c)$ take the form:

| | | |
|---|---|---|
| 8 or 9 | $x$ | $a$ |
| 17 or 18 | $y$ | $b$ |
| 17 or 18 | $z$ | $c$ |

where $x \geq y \geq z$, $x + y + z = 44$, and $a + b + c = 24$.

Recall that $f_h = x \cdot a + y \cdot c + z \cdot b$ is the number of phantom points in the 8- or 9-point flat. Then $f_h \geq 381$. Since the 44-cap is one point taken away from a 45-cap, it can only have divisions 15-15-14, 18-17-9, and 18-18-8.

If $x = 15$, $y = 15$, and $z = 14$, then $f_h \leq 15 \cdot 24 = 360$, which contradicts the initial assumption.

If $x = 18$ and $y \geq 17$, then $a \leq 9$, $b \leq 9$, and $c \leq 9$, for otherwise either a complete 44-cap or an inadmissable cap is created.

Maximizing $a$ and $c$ maximizes $f_h$, and so $f_h \leq 9 \cdot x + 9 \cdot y + 6 \cdot z = 9 \cdot 44 - 3 \cdot z = 396 - 3 \cdot z$. However, $z \geq 8$, and so $f_h \leq 372$, again contradicting the initial assumption.

Therefore, if there is a 44-44-24 division of a 112-cap in AG(6,3), there is a complete 44-cap in AG(5,3). □

If there are no complete 44-caps, we have eliminated the divisions 45-44-23, 45-43-24, and 44-44-24 for the 112-caps in AG(6,3).

**Corollary 12** *If there are no complete* 44*-caps in AG*(5, 3)*, then any* 112*-cap in AG*(6, 3) *must have at least one* 45-45-22 *division.*

## 4 Study of 44-caps in AG(5,3)

**Theorem 13** *No* 44*-caps in AG*(5, 3) *have a* 19*- or* 20*-point hyperplane.*

*Proof* We begin by recalling some facts about caps in AG(4,3). For the 20-cap, which is unique except for affine transformations, there are 30 8-6-6 divisions and 10 9-9-2 divisions. Each point of the 20-cap appears in 1 of the 10 2-point hyperplane cosets. There is one exterior point $c$ which appears in all 10 of the 2-point hyperplane cosets. All other exterior points appear in 4 of the 10 2-point hyperplane cosets.

All 19-caps in AG(4,3) can be extended to 20-caps [6], and the 19-cap is unique up to affine transformations. Thus, the 19-cap has 1 9-9-1 division, 9 9-8-2 divisions, 12 7-6-6 divisions, and 18 8-6-5 divisions. There is one exterior point $c$ which appears in all 10 of the 1 or 2-point hyperplane cosets and one exterior point $p$ (which can be made a point to complete the cap) which appears in 1 of the 1- or 2-point hyperplane cosets. All other exterior points appear in 4 of the 10 1- or 2-point hyperplane cosets.

For a 20-$m$-$r$ division, there must be at least one intersection square having a 9-9-2 division for the 20-point hyperplane coset such that at least $\frac{4}{10}mr$ phantom points are in the 2-point solid of the 20-point hyperplane, as there are $m \cdot r$ phantom points.

For a 19-$m$-$r$ division, the situation is more complex. We will find a requirement for the 9-8-2 divisions by first searching for the maximum number of phantom points that can be in the 1-point solid of the 19-point hyperplane. Let us call this max. To maximize the sum of the $f_i$, we take max as the $f_i$ for the 9-9-1 subdivision and take the maximum $f_i$ from the 9-8-2 subdivisions and assume that all $f_i$ from the 9-8-2 subdivisions have this value. To minimize the required sum, we assume as many phantom points as possible are equal to $p$ and no phantom points are equal to $c$. Each phantom point that is equal to $p$ only appears in a 1- or 2-point solid of the 19-point hyperplane once instead of 4 times, reducing the required sum by 3 times the number of phantom points equal to $p$. To meet the requirement, there must be at least one intersection square $i$ such that $f_i$, the number of phantom points in the 2-point solid of the 19-point hyperplane, satisfies: $9 \cdot f_i + max \geq 4 \cdot m \cdot r - 3 \cdot maxphantom$, where maxphantom is the maximum number of phantom points equal to $p$.

We will now use a computer search to find intersection squares satisfying these conditions.

The search will go through the possible divisions in the following order:

check (20,20,4).
check (20,19,5).
for ($n = 6$; $n \leq 12$; $n++$)

(20, $24 - n$, $n$).
(19, $25 - n$, $n$).

For every division, for every intersection square that is found that satisfies the requirement, we will later exclude that particular intersection square via an independent method. This then together completes the proof that the division in question is impossible.

So since this division does not exist, it cannot occur in any intersection square describing a valid 44-cap. We therefore excluded the intersection squares containing that division from future searches, and optimized in this way the search through the possible intersection squares.

Finally, two last tricks that were used in the computer search. Firstly, the following looks plausible, but is actually impossible:

| | | |
|---|---|---|
| 2 | ? | ? |
| 8 | 6 | 5 or 6 |
| 9 | ? | ? |

The reason is that the 8 in 9-8-2, which can be extended to a 9-cap in AG(3,3), is different from the 8 in 6-6-8, which is complete. The easiest way to show this is to note that in the 9-cap in AG(3,3), adding any other point will create exactly two lines. However, for the 20-cap, there is one point $c$ such that adding it creates 10 lines. For all 6-6-8 divisions, $c$ is in the 8-point hyperplane coset, and adding it to the cap creates 4 lines within that hyperplane coset.

Secondly, for a 19-$m$-$r$ division, a starting condition is $maxphantom \leq m$ and $maxphantom \leq r$. This can sometimes be improved depending on the intersection square. For example, for:

| | | |
|---|---|---|
| 2 | 7 | 6 |
| 8 | 3 | 2 |
| 9 | 4 | 3 |

**Table 1** Intersection squares eliminated by direct computer search

| | | |
|---|---|---|
| 2 | 9 | 2 |
| 9 | 9 | 0 |
| 9 | 2 | 2 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 9 | 4 | 0 |
| 9 | 5 | 2 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 9 | 5 | 0 |
| 9 | 4 | 2 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 8 | 5 | 2 |
| 9 | 4 | 1 |

| | | |
|---|---|---|
| 2 | 9 | 5 |
| 8 | 4 | 1 |
| 9 | 5 | 1 |

| | | |
|---|---|---|
| 2 | 9 | 5 |
| 8 | 4 | 0 |
| 9 | 5 | 2 |

| | | |
|---|---|---|
| 2 | 8 | 5 |
| 8 | 4 | 1 |
| 9 | 5 | 2 |

**Table 2** Other intersection squares eliminated by direct computer search

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 9 | 5 | 1 |
| 9 | 4 | 1 |

| | | |
|---|---|---|
| 2 | 9 | 5 |
| 9 | 4 | 1 |
| 9 | 4 | 1 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 9 | 4 | 2 |
| 9 | 4 | 1 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 8 | 5 | 1 |
| 9 | 4 | 2 |

| | | |
|---|---|---|
| 2 | 9 | 4 |
| 8 | 4 | 1 |
| 9 | 5 | 2 |

$maxphantom \leq 9$ because there are at most 3 phantom points equal to $p$ involving the top middle solid and lower right solid, there are at most 2 phantom points equal to $p$ involving the center solid and middle right solid, and there are at most 4 phantom points equal to $p$ involving the bottom middle solid and upper right solid. This improves the bound of $maxphantom \leq 11$ which comes from the division alone.

From the computer search, up to affine equivalence, we found that a 44-cap must contain one of the following intersection squares:

We used a computer search looking for the actual caps to exclude the intersection squares in Tables 1 and 2. These searches started with the 19 or 20 point hyperplane on the left, then proceeded to try and fill in the points in the remaining solids in the following order: right bottom, middle top, right middle, right top, center, middle bottom. □

**Theorem 14** *There are no complete* 44*-caps in* $AG(5, 3)$.

The proof proceeds in a number of steps.

We have excluded the possibility of a 44-cap which has a 19- or 20-point hyperplane. Therefore, any 44-cap in AG(5,3) can have at most 18-hyperplanes. The possible divisions are 18-18-8, 18-17-9, 18-16-10, 18-15-11, 18-14-12, 18-13-13, 17-17-10, 17-16-11, 17-15-12, 17-14-13, 16-16-12, 16-15-13, 16-14-14, and 15-15-14.

Consider Lemmas 2 and 3. These equations give the average value of $N_2(h)$ and $N_3(h)$. For each division, look at how much these values differ from the averages.

The results are:

| Division | $121(N_3(h) - N_{3_{avg}})$ | $121(N_2(h) - N_{2_{avg}})$ |
|---|---|---|
| 18-18-8 | 32076 | 2574 |
| 15-15-14 | −18018 | −1419 |
| 18-17-9 | 19008 | 1485 |
| 18-16-10 | 8844 | 638 |
| 18-15-11 | 1584 | 33 |
| 18-14-12 | −2772 | −330 |
| 18-13-13 | −4224 | −451 |
| 17-17-10 | 6908 | 517 |
| 17-16-11 | −2167 | −209 |
| 17-15-12 | −8217 | −693 |
| 17-14-13 | −11242 | −935 |
| 16-16-12 | −10032 | −814 |
| 16-15-3 | −14751 | −1177 |
| 16-14-14 | −16324 | −1298 |

Now represent groups of divisions as vectors in $\mathbb{R}_{14}$. So $(x_1, x_2, ..., x_{14})$ represents $x_1$ 18-18-8 divisions, $x_2$ 15-15-14 divisions, $x_3$ 18-17-9 divisions, etc.

For a vector to satisfy Lemmas 2 and 3, the following two equations must be satisfied:

$$32076 \cdot x_1 - 18018 \cdot x_2 + 19008 \cdot x_3 + \cdots = 0 \quad \text{and}$$
$$2574 \cdot x_1 - 1419 \cdot x_2 + 1485 \cdot x_3 + 638 \cdot x_4 + \cdots = 0.$$

We want a basis of vectors which satisfy these two equations. One such basis is as follows:

| Vector | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $x_7$ | $x_8$ | $x_9$ | $x_{10}$ | $x_{11}$ | $x_{12}$ | $x_{13}$ | $x_{14}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 6 | 4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 11 | 24 | 0 | 9 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 23 | 42 | 0 | 0 | 12 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 7 | 12 | 0 | 0 | 0 | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 89 | 150 | 0 | 0 | 0 | 0 | 36 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 61 | 150 | 0 | 0 | 0 | 0 | 0 | 108 | 0 | 0 | 0 | 0 | 0 | 0 |
| 7 | 173 | 282 | 0 | 0 | 0 | 0 | 0 | 0 | 216 | 0 | 0 | 0 | 0 | 0 |
| 8 | 23 | 30 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 24 | 0 | 0 | 0 | 0 |
| 9 | 28 | 33 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 27 | 0 | 0 | 0 |
| 10 | 3 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 0 | 0 |
| 11 | 7 | −18 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 72 | 0 |
| 12 | 7 | −12 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 27 |

If there is a vector which satisfies Lemmas 2 and 3, it must be a linear combination of these vectors.

For a cap in AG(5, 3), there is also the condition that $\sum_{i=1}^{14} x_i = 121$, as there are 121 divisions. This can be used to reduce the space of solutions to an 11-dimensional vector space. However, this does not provide a substantial simplification.

We also have the condition that there must be a non-negative integer of each division, but this still leaves a large number of possibilities. We need another condition.

To establish this condition, let us look at weighted sets. For now, instead of each point having a weight of either 0 (point not in the set) or 1 (point in the set), let us allow each point to have any weight that is a non-negative integer.

Here, the total number of points is now the sum of all these weights, and each division is now determined by the sum of the weights within each flat.

**Lemma 15** *For any such weighted set, if the total number of points is* $1 \mod 3$*, there exists a point such that adding 2 to its value creates a new weighted set such that for any division x-y-z,* $x = y = z \mod 3$.

*Similarly, if the total number of points is* $2 \mod 3$*, there exists a point such that adding 1 to its value creates a new weighted set such that for any division x-y-z,* $x = y = z \mod 3$.

*Proof* Look at the points as columns of a matrix with multiplicities. The condition that for any division $x$-$y$-$z$, $x = y = z \mod 3$ is equivalent to the condition that for all vectors $v$, if $x$ is the number of columns $c$ such that $v \cdot c = 0$, $y$ is the number of columns $c$ such that $v \cdot c = 1$, and $z$ is the number of columns $c$ such that $v \cdot c = 2$, then $x = y = z \mod 3$.

Let $s$ be the sum of all the columns. Then $s \cdot v = y + 2z$. If there are $3n$ points, $x + y + z = 0 \mod 3$, so $x = y = z \mod 3$ if and only if $y = z \mod 3$ if and only if $s \cdot v = y + 2z = 0 \mod 3$.

If $n = 1 \mod 3$, we add twice the column $s$ to the weighted set. If $n = 2 \mod 3$, we add once the column $-s$ to the weighted set.

Hence, we can always choose the point so that adding it (or adding it twice) makes the sum $s' = 0 \mod 3$ for the new weighted set. As shown above, this is sufficient to guarantee that for any division $x'$-$y'$-$z'$ for the new weighted set, $x' = y' = z' \mod 3$.

This completes the proof. □

Since this is true for weighted sets, it is true for caps as well. Thus, for any possible 44-cap in AG(5,3), there exists a point such that adding 1 to its weight creates a weighted set (not necessarily a cap) such that for any of its divisions $x$-$y$-$z$, $x = y = z \mod 3$.

Let us call this point $c$. For each possible division, let us see where $c$ must be.

For 18-18-8, $c$ is in the 8-point hyperplane.
For 15-15-14, $c$ is in the 14-point hyperplane.
For 18-17-9, $c$ is in the 17-point hyperplane.
For 18-16-10, $c$ is in the 18-point hyperplane.
For 18-15-11, $c$ is in the 11-point hyperplane.
For 18-14-12, $c$ is in the 14-point hyperplane.
For 18-13-13, $c$ is in the 18-point hyperplane.
For 17-17-10, $c$ is in the 10-point hyperplane.
For 17-16-11, $c$ is in the 16-point hyperplane.
For 17-15-12, $c$ is in the 17-point hyperplane.
For 17-14-13, $c$ is in the 13-point hyperplane.
For 16-16-12, $c$ is in the 12-point hyperplane.
For 16-15-13, $c$ is in the 15-point hyperplane.
For 16-14-14, $c$ is in the 16-point hyperplane.

Now let us combine this with Lemma 5.

Applying it to a 44-cap in AG(5,3), if $c$ is not in the 44-cap, $s = 1760$ and the average number of points of the cap that are in the same hyperplane as $c$ is $\frac{1760}{121} = \frac{160}{11}$. If $c$ is in the cap, $s = 1841$ and the average number of points of the 44-cap that are in the same hyperplane as $c$ is $\frac{1841}{121}$. Either way, the average number of points that are in the same hyperplane as $c$ is at least $\frac{1760}{121} = \frac{160}{11}$. Let us calculate the averages for each of the basis vectors we constructed above.

| Division | num of this division | num of 18-18-8 | num of 15-15-14 | Average |
|---|---|---|---|---|
| 18-17-9 | 4 | 1 | 6 | $\frac{160}{11}$ |
| 18-16-10 | 9 | 11 | 24 | $\frac{293}{22}$ |
| 18-15-11 | 12 | 23 | 42 | $\frac{904}{77}$ |
| 18-14-12 | 3 | 7 | 12 | $\frac{133}{11}$ |
| 18-13-13 | 36 | 89 | 150 | $\frac{692}{55}$ |
| 17-17-10 | 108 | 61 | 150 | $\frac{3688}{319}$ |
| 17-16-11 | 216 | 173 | 282 | $\frac{8788}{671}$ |
| 17-15-12 | 24 | 23 | 30 | $\frac{92}{7}$ |
| 17-14-13 | 27 | 28 | 33 | $\frac{1037}{88}$ |
| 16-16-12 | 6 | 3 | 2 | $\frac{124}{11}$ |
| 16-15-13 | 72 | 7 | $-18$ | $\frac{92}{7}$ |
| 16-14-14 | 27 | 7 | $-12$ | $\frac{160}{11}$ |

All of these values are less than $\frac{160}{11}$ except for the group with 18-17-9 and the group with 16-14-14. Note that there cannot be a negative number of any of these basis vectors because that would imply a negative number of some division. Since no value is greater than $\frac{160}{11}$, the only way to get a value of $\frac{160}{11}$ is to have only the 1st and 12th basis vectors involving 18-17-9 divisions and 16-14-14 divisions, respectively, and it is impossible to have a value greater than $\frac{160}{11}$.

Thus, the only possible divisions for a 44-cap are 18-18-8, 18-17-9, 15-15-14, and 16-14-14. Also, since the average number of points that are in the same hyperplane as $c$ must be $\frac{160}{11}$, $c$ must be an exterior point to the 44-cap.

Now let us look at intersection squares of an $n$-cap in $k$ dimensions. For an intersection square, each letter represents a $(k-2)$-dimensional space.

| | | |
|---|---|---|
| $a$ | $d$ | $g$ |
| $b$ | $e$ | $h$ |
| $c$ | $f$ | $i$ |

Let us look at the hyperplanes that contain the $(k-2)$-dimensional space with $a$ points. (A similar result is true for all of the other such spaces.)

These hyperplanes have $a+d+g$, $a+e+i$, $a+h+f$, and $a+b+c$ points, respectively. Let us call the sum $s$.

$$s = 4\cdot a + b + c + d + e + f + g + h + i = 3\cdot a + n.$$
$$s \mod 3 = n \mod 3.$$

Let us apply this to an intersection square of a 44-cap. One of the nine $(k-2)$-dimensional spaces will contain the point $c$ which extends the 44-cap to a set of 45 points such that for each division $x$-$y$-$z$, $x = y = z \mod 3$. This point c can only be contained in the 8-point hyperplane of 18-18-8 ($-1 \mod 3$), the 17 point hyperplane of 18-17-9 ($-1 \mod 3$), the 14-point hyperplane of 15-15-14 ($-1 \mod 3$), or the 16-point hyperplane of 16-14-14 ($1 \mod 3$). Let $u$ be the number of 16-14-14 divisions in a particular intersection square. Let $s$ be the sum of the hyperplane cosets containing the $(k-2)$-dimensional space containing $c$.

$$\begin{aligned}
&s \mod 3 = n \mod 3 = -1 \mod 3.\\
&(u \cdot 1 + (4-u) \cdot (-1)) \mod 3 = -1 \mod 3.\\
&2 \cdot u \mod 3 = 3 \mod 3 = 0 \mod 3.\\
&u = 0 \text{ or } u = 3.
\end{aligned}$$

Each division appears in 40 intersection squares, as if we extend AG(5, 3) to PG(5, 3), there is a one-to-one correspondence between planes at infinity and intersection squares, and each solid at infinity contains 40 such planes. Any pair of divisions appears together exactly once in an intersection square, as two solids at infinity intersect in a unique plane at infinity.

Thus, if there is a 16-14-14 division, there must be at least 81 16-14-14 divisions.

Assume that there is a 16-14-14 division. Looking at the basis vectors, only $(7, -12, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 27)$ and $(1, 6, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ can be part of the cap. Let $x$ be the number of the first vector and let $y$ be the number of the second vector. We have $22x + 11y = 121$ (to have 121 divisions in total), and $x \geq 3$ (to have at least 81 16-14-14 divisions). The number of 15-15-14 divisions is $-12x + 6y$. This must be non-negative, so $y \geq 6$. But then $22x + 11y > 121$; a contradiction. Thus, there are no 16-14-14 divisions.

We conclude that $x = 0$, so $y = 11$.

This implies that any 44-cap in AG$(5, 3)$ must have 44 18-17-9 divisions, 11 18-18-8 divisions, and 66 15-15-14 divisions. Furthermore, $c$ must be an exterior point, and adding it to the cap must make the divisions 55 18-18-9 divisions and 66 15-15-15 divisions. This satisfies the conditions for the pair count and the triple count and must therefore be a cap. Thus, there are no complete 44-caps in AG(5,3).

This concludes the proof of Theorem 14.

## 5 Uniqueness of the 112-cap in AG$(6, 3)$

Thus, every 112-cap in AG$(6, 3)$ admits a 45-45-22 division.

We will now prove that every 112-cap in AG$(6, 3)$ is equal to, in the projective closure PG(6, 3) of AG(6, 3), the set of points on a cone with vertex a point $c$ and base a 56-cap in a hyperplane of PG(6, 3) not through $c$, except for the vertex $c$ and the points of the cone in one hyperplane of PG(6, 3) not through $c$.

**Theorem 16** *The maximal size of a cap in AG*$(6, 3)$ *is* 112*, and all the* 112*-caps in AG*$(6, 3)$ *are affinely equivalent.*

*Every* 112*-cap in AG*$(6, 3)$ *is equal to a cone with vertex c and base a* 56*-cap in PG*$(5, 3)$*, except for the vertex c and for the points of the cone in one hyperplane not through c.*

Consider a 45-45-22 division of a 112-cap in AG(6,3).

From Lemma 6, $\sum_h f_h = 20mr$, where $f_h$ is the number of phantom points in a 9-point flat of the 45-cap on the left. Furthermore, $m = 45$ and $r = 22$.

For a given intersection square, affine transform it to look like this, where $x \geq y \geq z$ (similar to what was done in examining the possibility of a 45-44-23 division):

| | | |
|---|---|---|
| 9 | $x$ | $a$ |
| 18 | $y$ | $b$ |
| 18 | $z$ | $c$ |

Since the middle hyperplane is a 45-cap, either $x = y = z = 15$ or $x = y = 18$ and $z = 9$.

If $x = y = z = 15$, $f_h = 15 \cdot 22 = 330$.

If $x = y = 18$ and $z = 9$, we maximize $f_h$ by maximizing $a$ and $c$. However, $a$, $b$, and $c$ must all be smaller than or equal to 9 as otherwise a cap in AG(5, 3) with at least 46 points is created. Thus, $f_h$ is maximal when $a = c = 9$ and $b = 4$.

This gives $f_h = 18 \cdot 9 + 18 \cdot 9 + 9 \cdot 4 = 360$.

No other possibilities give this value.

If we assume $f_h$ attains this maximum for all 55 subdivisions where the left hyperplane is divided 18-18-9, we get $\sum_h f_h = 55 \cdot 360 = 19800$. The required value is $20 \cdot m \cdot r = 20 \cdot 45 \cdot 22 = 19800$. Thus, $f_h$ must obtain this maximum for all 55 subdivisions. If the left hyperplane coset is 18-18-9, the intersection square must look like this (or an equivalent intersection square):

| | | |
|---|---|---|
| 9 | 18 | 9 |
| 18 | 18 | 4 |
| 18 | 9 | 9 |

Now let us use phantom points in a slightly different way. Pick a point $c_1$ of the 112-cap in the left hyperplane coset and a point $c_2$ of the 112-cap in the middle hyperplane coset. For each such pair of points, call $-c_1 - c_2$, the point needed to complete the line, a phantom point. Each phantom point is in the right hyperplane coset, although multiple phantom points might be equal to the same point.

Now pick a point $p$ in the right hyperplane. For each subdivision, define $f_h$ to be the number of phantom points in the same 4-dimensional space as $p$.

**Lemma 17** $\sum_h f_h = 40 \cdot 45 \cdot 45 + 81 \cdot x$, *where $x$ is the number of phantom points equal to $p$ and where the sum is taken over all subdivisions, not just the ones where the 45-point hyperplanes are 18-18-9.*

*Proof* If a phantom point is equal to $p$, it will be in the same 4-dimensional space as $p$ for all 121 subdivisions. Otherwise, it will be in the same 4-dimensional space as $p$ for 40 out of the 121 subdivisions. There are $45 \cdot 45$ phantom points, so this can be expressed as:

$$\sum_h f_h = 40 \cdot (45 \cdot 45 - x) + 121 \cdot x = 81000 + 81 \cdot x.$$

□

If the 45-point hyperplane cosets are 15-15-15, $f_h = 15 \cdot 45 = 675$. If the 45-point hyperplane cosets are 18-18-9, if $p$ is in the 4 point 4-dimensional space, $f_h = 2 \cdot 18 \cdot 18 + 9 \cdot 9 = 729$. Otherwise, $f_h = 18 \cdot 18 + 2 \cdot 9 \cdot 18 = 648$.

Looking at just the 55 subdivisions where the 45-point hyperplane cosets are 18-18-9, let $n$ be the number of times $p$ is in the 4-point 4-dimensional space.

$$\begin{aligned}
&\sum_h f_h = 66 \cdot 15 \cdot 45 + (55 - n) \cdot 648 + n \cdot 729 = 80190 + 81 \cdot n, \\
&80190 + 81 \cdot n = 81000 + 81 \cdot x, \text{ so} \\
&n = 10 + x.
\end{aligned}$$

Now, from Lemma 15, there must be a point $c$ in the 22-point hyperplane coset such that adding 2 to its weight creates an arrangement such that for all divisions $x'$-$y'$-$z'$, $x' = y' = z'$ mod 3.

For all 9-9-4 divisions of the 22-cap, $c$ must be in the 4-point hyperplane.

Thus, for $c$, in the equation above, $n = 55$. So $x = 45$.

There are 45 phantom points equal to $c$. This means that if a point $p$ in the left hyperplane coset is in the 112-cap, the point $-c - p$, which is in the middle hyperplane coset, must also be in the 112-cap. Thus, the left hyperplane and $c$ completely determine the middle hyperplane.

Alternatively, we see that the 112-cap is starting to get the structure of a cone. As the following arguments will show, in PG(6, 3), $c$ is in fact the vertex of a cone with base the 56-cap in PG(5, 3). This 112-cap in AG(6, 3) will consist of all points on this cone, except for the vertex $c$ and for the points in one hyperplane not through $c$. This will lead to the uniqueness of the 112-cap in AG(6, 3).

Pick two points $p_1$ and $p_2$ in the left 45-point hyperplane coset. Extend this to PG(5, 3); we will also use the hyperplane at infinity of AG(5, 3) in the remaining part of these arguments. Let $a$ be the point at infinity they are collinear with.

Now look at the phantom points from $p_1, -c - p_2$ and $p_2, -c - p_1$. They are $c + p_1 - p_2$ and $c - p_1 + p_2$. These are the two points collinear with $a$ and $c$.

Note that every phantom point can be described in this way. Thus, for a point $p$ in the right hyperplane coset, if $a$ is the point at infinity collinear with $p$ and $c$, $p$ has no phantom points equal to it if and only if $a$ is not collinear with any pair of points in the left hyperplane coset.

From [6], it follows that there are 11 such points $a$ in the hyperplane at infinity of AG(5, 3). For such a point $a$ in the hyperplane at infinity of AG(5,3), there are exactly 2 points collinear with $a$ and $c$. This gives exactly 22 points with no phantom points equal to them, so the remaining 22 points of the 112-cap in AG(6, 3) are also determined. Thus, for any 112-cap in AG(6, 3), the remaining points are completely determined. We have proven that this 112-cap in AG(6, 3) is a cone with vertex $c$ and base a 56-cap in PG(5, 3), except for the vertex $c$ and for the points of the cone in one hyperplane not through $c$.

The preceding result shows that the 112-cap in AG(6,3) is unique.

Note that this 112-cap must be complete because there is no possible place for an additional point. We have just shown that there is no place for it in the 22-point hyperplane, and clearly there is no place for it in either of the 45-point hyperplanes, as these are complete caps in AG(5, 3).

Since this 112-cap is complete, this also proves that the maximal size of a cap in AG(6, 3) is 112, and all the 112-caps in AG(6, 3) are affinely equivalent.

This leads to Theorem 16, the principal result of this article.

**Acknowledgements** The author wishes to thank Prof. Dr. Robert Calderbank, Princeton University, and Prof. Dr. Leo Storme, Ghent University, for extensive help in editing the article. He also wishes to thank Dr. Jan De Beule, Ghent University, for the hospitality during his stay at the Department of Pure Mathematics and Computer Algebra, Ghent University.

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